

Runze Ma

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Education

Monash University Malaysia	Master of Artificial Intelligence	2024.07 – 2026.11
Huazhong University of Science and Technology	B.Eng. in Automation	2020.09 – 2024.06

Publications

TRACE: Topology-Regularized Autoencoder for Interpretable Clinical ECG Diagnosis

- *NeurIPS 2026, Under Review, First Author (Co-Author)*
- We propose TRACE, a topology-regularized autoencoder that disentangles latent semantics into clinically meaningful subspaces via physiological constraints, achieving interpretable and robust ECG diagnosis.

LiteMedCoT-VL: Parameter-Efficient Adaptation for Medical Visual Question Answering [\[link\]](#)

- *NLPCC 2026, Under Review, First Author*
- We propose LiteMedCoT-VL, a 2B vision-language model achieves 64.9% on medical VQA by distilling chain-of-thought reasoning from a 235B teacher via LoRA, outperforming models with twice the parameters.

BEAT-Net: Injecting Biomimetic Spatio-Temporal Priors for Interpretable ECG Diagnosis [\[link\]](#)

- *IEEE Journal of Biomedical and Health Informatics, Under Review, First Author*
- We introduce BEAT-Net, a biomimetic framework mimicking cardiologist reasoning through QRS-centered tokenization and hierarchical spatio-temporal encoders for efficient, interpretable ECG diagnosis.

AEGIS-NET: Exploiting Physio-Kinetic Coupling Via Asymmetric Task-Aware Gating For Unified Bio-metrics, Activity Sensing, And Health Profiling [\[link\]](#)

- *Knowledge-Based Systems, Under Review, Second Author*
- We propose AEGIS-Net, a unified multi-modal framework for edge IoT devices that synergizes Electrocardiography, Photoplethysmography, and Inertial Measurement Unit signals via an Asymmetric Task-Aware Gating mechanism.

Stacking-Enhanced Bagging Ensemble Learning for Breast Cancer Classification with CNN [\[link\]](#)

- *ICEEM 2023, Second Author*
- We present BSECNN, a CNN-based ensemble using Bagging and Stacking for breast cancer classification.

REALM: Retrospective Encoder Alignment for LFP Modeling [\[link\]](#)

- *Journal of Neural Engineering, Under Review, Third Author*
- We propose REALM, a retrospective distillation framework that enables high-performance causal LFP decoding.

CPR: Causal Physiological Representation Learning for Robust ECG Analysis under Distribution Shifts [\[link\]](#)

- *NeurIPS 2026, Under Review, Fifth Author*
- We propose CPR, a robust diagnostic framework leveraging structural causal modeling to disentangle invariant pathological morphology from adversarial artifacts, achieving certified-level robustness and real-time inference efficiency.

Work Experience

NeuroML Lab at Tsinghua University	Research Assistant	2026.05 – Present
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- Developed and trained foundation models for biosignals.

Laboratory of Advanced Bioelectronics at SUAT [link]	Research Assistant	2025.11 – Present
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- Developed and trained multimodal pre-training models for biosignals.

Kingda Education	AI Technical Consultant	2025.10 – Present
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- Delivered workplace AI training and provided technical consulting support for practical AI adoption.
- Designed and implemented task-oriented AI agents to streamline internal workflows.

Antalya Technology [link]	AI Product & Operations Lead	2024.03 – Present
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- Designed core prompts for **IELTS Master** [\[link\]](#), constructing a structured CoT framework to improve stability.
- Extracted user data via SQL and built automated Power BI dashboards to drive product iteration.

Juxin Financial Consulting	Data Analysis Intern	2024.11 – 2025.02
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- Integrated OCR with LLM APIs to automate invoice processing, boosting reconciliation efficiency by 50%.

Project Experience

-  **Recipe Generation: From RNNs to RAG Menu Designer** 2026.04 – 2026.05
- Trained encoder-decoder RNNs with Bahdanau attention from scratch for ingredient-to-recipe generation.
 - Fine-tuned T5 and GPT-2 with LoRA, and built a Menu Designer with RAG pipeline and LLM-as-Judge evaluation.
-  **Multi-Agent RL for Autonomous Shuttle Delivery** 2025.05 – 2025.06
- Designed a tabular Q-learning MARL system on a 5×5 grid with collision detection and hybrid reward shaping.
 - All four agents achieved 100% delivery success rate in focal agent evaluation across 16 start configurations.
-  **Pac-Man AI: Classical Search & Adversarial Planning** 2025.03 – 2025.04
- Implemented A* search with domain-specific heuristics for single-goal, multi-corner, and full food collection pathfinding.
 - Designed alpha-beta pruning with a handcrafted evaluation function for adversarial ghost avoidance.
-  **Drone-Based Dual-Modal Vehicle Detection** 2023.11 – 2024.06
- Improved YOLO with RGB-IR feature fusion to resolve detection failures in complex lighting and weather.
 - Validated on DroneVehicle dataset and optimized quantization for real-time embedded deployment.
- 2023 Summer Ethics of AI Program at North Carolina State University [\[link\]](#)** 2023.07 – 2023.08
- Surveyed AI ethical controversies and built a multi-dimensional assessment framework.
 - Authored a report proposing targeted ethical guidelines, receiving positive mentor feedback.
-  **Pain Intensity Estimation via Spatio-Temporal Networks** 2023.02 – 2023.07
- Optimized feature extraction on UNBC-McMaster dataset by comparing multiple spatio-temporal architectures.
 - Validated hybrid model advantages for non-invasive pain assessment via cross-validation.
- Ensemble Learning with VAE for Breast Cancer Classification** 2022.11 – 2023.02
- Proposed a Bagging & Stacking framework to address sample imbalance and improve reliability.
 - Integrated VAE for data augmentation and feature selection to optimize classification performance.

Certificates & Honors

2025 Smart Wearable and Sports Health Technology Challenge - <i>National 1st Place</i>	2026.03
Technology Innovation Scholarship - <i>1/30</i>	2022.09
Certificate of Professional Knowledge Assessment in Mathematical Modeling	2022.08
Certificate of the University Artificial Intelligence Training Camp [link]	2022.08
MathorCup College Math Modeling Challenge [link] - <i>First Prize (Top 5%)</i>	2022.05
Mathematical Contest in Modeling - <i>Honorable Mention (Top 20%)</i>	2022.02
Social Welfare Scholarship - <i>1/30</i>	2021.09

Skills

- **Programming:** Python > C = Matlab > Java > C++
- **Software:** Office, SQL, Power BI, VibeCoding
- **Languages:** Chinese (Native); English (IELTS 6.5); Arabic (Basic Spelling)